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// normal_app.c
#include <stdint.h>
#include <stddef.h>

enum {
    SVC_GET_VERSION = 0x01,
    SVC_SIGN_SHA256 = 0x02,
    SVC_VERIFY_X509_SUBJECT = 0x03,
};

static inline long smc_call(long a0, long a1, long a2, long a3, long a7) {
    register long ra0 asm("a0") = a0;
    register long ra1 asm("a1") = a1;
    register long ra2 asm("a2") = a2;
    register long ra3 asm("a3") = a3;
    register long ra7 asm("a7") = a7;
    asm volatile ("ecall"
        : "+r"(ra0)
        : "r"(ra1), "r"(ra2), "r"(ra3), "r"(ra7)
        : "memory");
    return ra0;
}

uint32_t get_monitor_version(void) {
    return (uint32_t)smc_call(0,0,0,0,SVC_GET_VERSION);
}

int sign_hash_demo(const uint8_t hash[32], uint8_t sig_out[64]) {
    return (int)smc_call((long)hash, (long)sig_out, 0, 0, SVC_SIGN_SHA256);
}

int verify_subject_demo(const uint8_t* cert_der, size_t cert_len) {
    return (int)smc_call((long)cert_der, (long)cert_len, 0, 0, SVC_VERIFY_X509_SUBJECT);
}

```